COM3105 Advanced Algorithms

Homework 3

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Question 1 *(4 points)*

Consider a stream $a_1, \ldots, a_m$ where each $a_i \in \{1, \ldots, n\}$.
Give a deterministic streaming algorithm that maintains the sum $a_1 + \ldots + a_m$ using $\mathcal{O}(\log m + \log n)$ space.

Answer

Algorithm:
Let S_i be the sum of the the stream at step i, for $i = 0, \ldots, m$.

- Initialise $S_0 = 0$.
- For each incoming element a_i, get the sum at time step i by $S_i = S_{i-1} + a_i$
- At any point, the current sum of the stream is given by S_i.
- After processing all m elements, $S_i = S_m$, i.e. the sum of all elements $a_1, \ldots, a_m$.

Space Complexity Analysis:
In the worst case, the value of any element a_i is n.
So we consider the following worst case run of the algorithm, where each $a_i = n$:

- Initially, $S_0 = 0$,
- $S_1 = S_0 + n = n$,
- $S_2 = S_1 + n = n + n = 2 \times n$,
- $\ldots$
- $S_i = S_{i-1} + n = i \times n$,
- $\ldots$
- $S_m = S_{m-1} + n = m \times n$

Therefore, the maximum value that S_i can take is $m \times n$.
To store $m \times n$, we need $\log(m \times n) = \log m + \log n$ bits.
Hence, the space complexity of the algorithm is $\mathcal{O}(\log m + \log n)$.

Question 2

(6 points)

Consider a stream $a_1, \ldots, a_m$ where each $a_i \in \{1, \ldots, n\}$.
Give a randomised streaming algorithm which approximates the sum $\sum a_1 + \ldots + a_m$ using $\mathcal{O}(\log\log m + \log n)$ space.
Give a rigorous analysis of the algorithm in terms of its quality and space complexity.

Answer

answer here
